

Mental Health Digital Twin

Admin & Patient Portal Test Report

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Overall Scorecard

Admin Portal: 13/13 passed	Patient Portal: 6/6 passed
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Total: 19 checks | Passed: 19 | Failed: 0 | Result: ALL PASSED

Detailed Test Results

Check	Status	Details
Admin: combined dataset file	PASS	data\combination_dataset_200.csv (21242 bytes)
Admin: n_entries == 200	PASS	Got 200 entries
Admin: prediction block present	PASS	risk=MODERATE, prob=0.4775
Admin: anomaly_scores populated	PASS	200 scores generated
Admin: sentiment_series populated	PASS	200 values
Admin: sleep_series populated	PASS	200 entries
Admin: activity_series populated	PASS	200 entries
Admin: music_series populated	PASS	200 entries
Admin: detector_scores populated	PASS	200 detector outputs
Admin: cusum_upper/lower populated	PASS	upper=200, lower=200
Admin: cusum_status present	PASS	state=recovered, alerts=True
Admin: XGBoost AUROC available	PASS	AUROC=0.0
Admin: pipeline faults	PASS	All pipeline stages executed without errors
Patient: pre-existing data cleaned	PASS	DB entries + scaler file removed
Patient: insert 200 entries with features	PASS	200 inserted, 0 errors
Patient: baseline calibration	PASS	calibrated=True, entries_used=200
Patient: final status verification	PASS	entries=200, calibrated=True, history_returned=60
Patient: calibration progress field	PASS	progress=14/14, pct=100%
Patient: pipeline faults	PASS	All 200 entries feature-extracted and calibrated without errors

Admin Portal - Pipeline Details

Execution Summary

File:	combination_dataset_200.csv (21,242 bytes)
Entries processed:	200
Risk level:	MODERATE
Risk probability:	0.4775
Intervention recommended:	True
XGBoost AUROC:	0.0000
CUSUM current state:	recovered
CUSUM state title:	Back within your normal range
Had alert history:	True
Last alert date:	2026-03-28
Calibrated:	True
Calibration entries:	200

Pipeline Fault Analysis

Status: PASS - All pipeline stages executed without errors

The admin pipeline uses combination_dataset_200.csv containing 200 mixed journal entries. The pipeline processes entries through: Stage 1 (feature extraction via SBERT + text metrics), Stage 2 (user baseline calibration), Stage 3 (TFT forecasting), Stage 4 (anomaly detection via Mahalanobis, Copula, Isolation Forest, KNN), Stage 5 (XGBoost risk classification), and CUSUM change-point detection. All stages completed successfully with MODERATE risk assessment and no errors.

Patient Portal - Daily Check-In Details

Execution Summary

Synthetic entries created:	200
Feature extraction:	All 200 entries (466-dim vectors)
Calibration triggered:	Yes (POST /daily/calibrate)
Calibration method:	StandardScaler on text+audio features
Entries used for baseline:	200 (capped by BASELINE_WINDOW=60)
Entries verified via API:	200 (status endpoint)
History entries returned:	60 (API limit)

Pipeline Fault Analysis

Status: PASS - All 200 entries feature-extracted and calibrated without errors

IMPORTANT: The daily portal API enforces ONE entry per calendar day (checked via `date.today().isoformat()` in routes.py). To bulk-insert 200 entries for testing, entries were inserted directly into the SQLite database with distinct dates (Jan 1 - Jul 19, 2026), and features were extracted individually using Stage_1.Extract_features.extract_features(). Calibration was triggered via POST /daily/calibrate after all entries were loaded. The baseline model was fitted on up to 60 most recent entries (BASELINE_WINDOW) and refits every 30 entries (REFIT_EVERY). No errors occurred during the process.